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SYNOPSIS

Project Title:

A Software License & Subscription Intelligence Platform

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1. ABSTRACT

Modern organizations use a large number of software applications and subscription-based services for development, design, communication, project management, cloud computing, and business operations. Managing these licenses manually can result in unused licenses, unnecessary expenditure, duplicate subscriptions, and missed renewal dates.

The proposed **Software License & Subscription Intelligence Platform** is a web-based system designed to provide centralized management of software licenses and subscriptions. The system maintains information about software products, vendors, license quantities, users, departments, costs, assignments, usage, and renewal dates.

The platform analyzes license utilization and identifies unused or underutilized licenses. It can generate recommendations for license reallocation, reduction, or renewal review. A centralized dashboard displays software expenditure, active licenses, utilization rates, upcoming renewals, and estimated savings.

The system combines web development, database management, data analytics, role-based access control, visualization, and a recommendation mechanism to improve software resource utilization and help organizations control subscription costs.

2. INTRODUCTION

The rapid adoption of cloud computing and Software-as-a-Service (SaaS) applications has increased the number of digital tools used by organizations. Companies may use multiple applications for communication, development, design, project management, security, productivity, and business operations.

Managing these subscriptions through spreadsheets and separate vendor portals makes it difficult to maintain accurate information about licenses and their actual usage. Organizations may continue paying for licenses that are rarely used or purchase new licenses even when unused licenses are already available.

Software license management therefore requires not only tracking what has been purchased, but also understanding how those licenses are being utilized. It is important to monitor license assignments, usage, costs, renewal dates, and subscription requirements.

The proposed **Software License & Subscription Intelligence Platform** provides a centralized interface to manage licenses, monitor usage, analyze costs, track renewals, and support better software-related decisions. The platform helps organizations identify unused resources and make informed decisions regarding license allocation, renewal, reduction, and purchase.

3. PROBLEM STATEMENT

Organizations commonly face difficulties in managing a large number of software licenses and subscriptions. Software information may be stored in different spreadsheets, vendor portals, and manually maintained records, making centralized monitoring difficult.

The major problems identified are:

1. Scattered Software and License Information

Software and license details may be maintained in different files or systems, making it difficult to access and manage information from one place.

2. Unused or Underutilized Licenses

Organizations may continue paying for licenses that are rarely used or not being used at all.

3. Unnecessary Subscription Expenditure

Without proper utilization and cost analysis, organizations may spend money on unnecessary licenses or subscriptions.

4. Duplicate Software Subscriptions

Different departments may purchase similar or duplicate software subscriptions without having proper visibility of existing resources.

5. Missed or Automatic Renewals

Manual tracking of renewal dates can result in missed renewals or unwanted automatic subscription renewals.

6. Limited Department-Wise Cost Visibility

Organizations may find it difficult to understand software expenditure across different departments.

7. Manual License Allocation and Approval

Manual allocation and approval processes can consume time and may result in inefficient license utilization.

8. Lack of Utilization Analytics

Basic records may show the number of purchased licenses but may not provide sufficient information about actual usage.

9. Difficulty Identifying Potential Savings

Without centralized cost and usage analysis, it becomes difficult to identify where software expenditure can be reduced.

Proposed Problem

Therefore, there is a need for a centralized platform that can manage software licenses and subscriptions, monitor their utilization, analyze costs, track renewals, and provide useful recommendations.

The proposed **Software License & Subscription Intelligence Platform** aims to address these limitations by providing centralized license management, usage analysis, cost analysis, renewal tracking, savings estimation, and recommendation features.

4. LITERATURE SURVEY AND EXISTING SYSTEM ANALYSIS

Existing Software Asset Management (SAM) and SaaS Management systems provide features such as software inventory, license tracking, user assignment, contract management, and expenditure monitoring.

Organizations may also manage software licenses using spreadsheets and manually maintained records. These approaches can store basic information but may provide limited automation and limited analysis of actual license utilization.

Traditional Software License Management can be represented as:

Software Purchase → License Record → Manual Assignment → Manual Monitoring → Renewal

The proposed intelligence-based approach can be represented as:

License Data → Usage Analysis → Cost Analysis → Renewal Analysis → Savings Analysis → Recommendation

Existing approaches may provide license tracking but may not always provide an integrated view of license utilization, expenditure, renewals, and potential savings.

Existing System Limitations

The major limitations identified in existing approaches are:

1. Limited centralized management of software licenses.
2. Dependence on spreadsheets and manual records.
3. Limited monitoring of actual software utilization.
4. Difficulty identifying unused and underutilized licenses.
5. Limited visibility of department-wise software expenditure.
6. Manual tracking of renewal dates.
7. Difficulty identifying duplicate subscriptions.
8. Limited cost-saving analysis.
9. Lack of recommendation-based decision support.

Proposed System Analysis

The proposed **Software License & Subscription Intelligence Platform** addresses these limitations by integrating license management with analytics and recommendation features.

The proposed system provides:

- Centralized software and license management
- User and department management
- License assignment and tracking
- Software usage monitoring
- License utilization analysis
- Cost and expenditure analysis
- Renewal tracking
- Duplicate subscription detection
- Savings estimation
- Software request and approval
- Recommendation generation
- Dashboard and reports
- Role-based access

The proposed approach can be represented as:

Software & License Data → Usage Analysis → Cost Analysis → Renewal Analysis → Risk/Savings Analysis → Intelligence Evaluation → Recommendation → Dashboard/Report

Thus, the proposed system aims to transform basic software license tracking into an intelligent decision-support platform for better software utilization and subscription cost management.

5. METHODOLOGY

The proposed **Software License & Subscription Intelligence Platform** follows a modular web-based architecture consisting of a frontend, backend server, relational database, analytics module, and recommendation engine.

The methodology consists of the following stages:

User Authentication

Administrators, managers, and employees access the system according to their assigned roles. Role-based access ensures that users can access only the functions permitted to them.

Software and License Management

The system maintains information about software products, vendors, license quantities, costs, assignments, subscription duration, and renewal dates.

License Usage Monitoring

The system records and analyzes license usage information to identify active, unused, and underutilized licenses.

Usage Analytics

The system calculates license utilization and analyzes usage information to identify licenses that require attention.

Cost Analysis

The system analyzes software expenditure and calculates potential savings based on unused or underutilized licenses.

Renewal Tracking

The system monitors license and subscription renewal dates and provides information about upcoming renewals.

Software Request and Approval

Employees can submit software requests. Managers or administrators can review and approve these requests based on organizational requirements and available licenses.

Recommendation Generation

The recommendation engine evaluates license usage, cost, renewal information, and predefined rules to generate recommendations for reallocation, reduction, renewal review, or use of existing licenses.

Dashboard and Reporting

The analyzed information is presented through dashboards and reports so that administrators and managers can monitor software usage, expenditure, renewals, and potential savings.

6. SYSTEM DESIGN METHODOLOGY

The system is designed using different functional layers that work together to provide centralized software license and subscription management.

6.1 Frontend Design

The frontend provides a user-friendly and responsive interface for administrators, managers, and employees.

It provides interfaces for:

- Login and authentication

- Dashboard
- Software management
- License management
- Employee management
- Usage analytics
- Cost analysis
- Renewal tracking
- Software requests
- Recommendations
- Reports

6.2 Backend Design

The backend handles the main application logic and communication between the frontend, database, analytics module, and recommendation engine.

It manages:

- Authentication and authorization
- Software and license records
- License assignment and reallocation
- Usage data
- Cost calculations
- Renewal monitoring
- Request approval
- Recommendation generation
- Report generation

6.3 Database Design

The relational database stores and organizes information required by the platform.

The database contains information related to:

- Users and employees
- Departments
- Software products
- Vendors
- Licenses

- License assignments
- Software usage
- Software requests
- Costs
- Renewal dates
- Recommendations

The database maintains relationships between users, departments, software, vendors, licenses, usage records, and requests. This allows the system to efficiently retrieve information for license tracking, analysis, reporting, and recommendation generation.

6.4 Analytics Module

The analytics module processes license and usage information to calculate utilization and identify unused or underutilized licenses.

The license utilization is calculated using:

License Utilization (%) = Active Licenses / Total Licenses × 100

For example, if 35 out of 50 licenses are actively used:

Utilization = 35 / 50 × 100 = 70%

The system can classify utilization as **Excellent, Good, Moderate, Low, or Critical** according to predefined thresholds.

The analytics module can also support cost analysis, renewal monitoring, savings estimation, and duplicate subscription detection.

6.5 Cost Analysis

The platform calculates software expenditure using:

Monthly Cost = Number of Licenses × Cost per License

For example, if 8 unused licenses cost ₹1,000 each per month:

Potential Monthly Saving = ₹8,000

Potential Annual Saving = ₹96,000

These values can be displayed on the dashboard to provide better visibility of software expenditure and potential savings.

6.6 Recommendation Engine

The recommendation engine evaluates license usage and predefined rules to generate useful recommendations.

For example:

If a license has not been used for more than 60 days and is not required, the system can recommend reviewing it for reallocation or reduction.

The system can also check whether an employee's software request can be fulfilled using an existing unused license before recommending a new purchase.

The recommendation engine helps organizations make better decisions regarding license allocation, renewal, reduction, and purchase.

User Flow

Login → Dashboard → Software/License Management → Usage & Cost Analysis → Renewal/Request Review → Recommendation → Dashboard/Report

7. OPERATIONAL FLOW

The existing 8-step flow is retained without changing the sequence.

Step 1 – Start

The administrator/user opens the web application.

Step 2 – Enter License & Subscription Details

The user enters the required software, license, subscription, cost, usage, and renewal information.

Step 3 – Submit for Analysis

The system validates the information and sends it to the analysis engine.

Step 4 – Perform Analysis

License Inventory → Usage Analysis → Cost Analysis → Renewal Analysis → Duplicate Detection → Risk Analysis → Savings Analysis

Step 5 – Intelligence Evaluation

The system analyzes the collected information and identifies important patterns, issues, and areas that require attention.

Step 6 – Score Calculation

The system calculates the overall license/subscription health score based on the analyzed information.

Step 7 – Recommendation

The system provides recommendations based on the analysis, such as license reallocation, reduction, renewal review, or using an existing unused license.

Step 8 – Report Generation

The complete analysis is stored and can be viewed through the dashboard or generated as a structured report.

8. MAIN FEATURES

User Authentication and Role Management

Provides secure login and role-based access for administrators, managers, and employees.

Software Management

Allows administrators to add, update, and manage software products and vendor information.

License Management

Maintains license details including quantity, assignment, cost, status, purchase information, and renewal dates.

User and Department Management

Maintains employee and department information and supports department-wise software management.

Usage Analytics

Analyzes software and license usage to identify active, unused, and underutilized licenses.

Cost and Savings Analysis

Provides information about software expenditure and estimates potential savings from unused or underutilized licenses.

Renewal Tracking

Tracks subscription and license renewal dates to help organizations review upcoming renewals.

Software Request and Approval

Allows employees to request software and enables authorized managers or administrators to review and approve requests.

Recommendation Engine

Generates recommendations based on license usage, cost, renewal information, and predefined conditions.

Dashboard and Reports

Provides a centralized dashboard showing important information such as active licenses, utilization rates, expenditure, upcoming renewals, recommendations, and estimated savings.

9. FUTURE SCOPE

The platform can be enhanced in the future by integrating:

1. Vendor APIs for automatic software usage collection.
2. Platforms such as GitHub, Microsoft 365, Jira, and Figma.
3. Machine learning for predicting future license requirements.
4. Automated email and renewal notifications.
5. AI-powered natural language queries.
6. Predictive software cost forecasting.
7. Automated license procurement recommendations.
8. Cloud deployment for multi-branch organizations.
9. Mobile applications for administrators and managers.
10. Single Sign-On (SSO) and advanced authentication.

These enhancements can improve automation, scalability, prediction, and decision-making capabilities of the platform.

10. CONCLUSION

The **Software License & Subscription Intelligence Platform** provides a centralized and intelligent approach to managing organizational software licenses and subscriptions.

The platform combines license management, usage analytics, cost tracking, renewal monitoring, role-based access, dashboards, reporting, and intelligent recommendations in a single system.

By identifying unused and underutilized licenses and analyzing software expenditure, the system can help organizations make informed decisions regarding license allocation, renewal, reduction, and potential cost savings.

The proposed project demonstrates the practical application of web development, database management, data analytics, authentication, visualization, and decision-support techniques to solve a real-world software license and subscription management problem.